CASE STUDY

St. Vincent Ferrer School Support System

Custom Helpdesk & Ticket Management Solution for Education

Project Scope

A complete IT helpdesk and ticket management system designed specifically for St. Vincent Ferrer Catholic School. The system streamlines how faculty and staff submit, track, and resolve technical support requests—replacing informal email chains and hallway conversations with an organized, priority-based workflow that ensures no issue falls through the cracks.

The Problem

Educational institutions face unique IT support challenges:

- **Disorganized requests:** Support requests came via email, phone calls, sticky notes, and hallway conversations—making it impossible to track or prioritize effectively
- **No priority visibility:** A teacher's projector failing mid-lesson received the same treatment as a routine password reset request
- **Communication gaps:** Staff had no way to check ticket status, leading to repeated follow-ups and frustration
- **Lost institutional knowledge:** Recurring issues were solved repeatedly because there was no documentation of past solutions

- **Resource allocation:** IT staff couldn't effectively plan their day without knowing what issues were pending

The Solution

A purpose-built ticketing system tailored to the school's specific needs:

- **Centralized submission portal:** One place for all IT requests with structured data capture
- **Visual priority system:** Color-coded urgency levels enable instant triage decisions
- **School-specific categories:** Issue types match actual school needs—projector setup, Google Classroom, Microsoft 365
- **Automatic notifications:** Email confirmations keep staff informed without manual follow-up
- **Audit trail:** Complete history of every ticket for accountability and pattern analysis

Key Features

Four-Tier Priority System

Color-coded urgency levels designed for at-a-glance triage, enabling IT staff to immediately identify critical issues:

Low	Normal	High	Critical
Not Urgent	Standard	Soon	Immediate

Structured Ticket Submission

Captures name, email, role, issue category, priority, and detailed description in a single form

School-Specific Categories

Hardware, software, presentation setup, Google Workspace, and Microsoft 365 support options

Email Notifications

Automatic confirmations keep submitters informed of ticket receipt and status changes

Accessibility Features

Full keyboard navigation and screen reader support for inclusive access

Custom School Branding

The interface was designed to reflect St. Vincent Ferrer's identity, creating a familiar and professional experience for staff. Custom dropdown selectors, buttons, and visual elements incorporate the school's official colors.

School Blue School Gold

Technical Architecture

Frontend

Backend API

- RESTful endpoints

Database

- Ticket ID generation

- Responsive HTML5/CSS3
- Vanilla JavaScript
- Custom form validation
- Accessible UI components

- Rate limiting protection
- Input sanitization
- Tenant identification

- Status tracking
- Assignment routing
- Resolution logging

Security & Reliability

- **Rate limiting:** Prevents abuse and ensures system availability during high-traffic periods
- **Input validation:** Server-side validation prevents malformed or malicious data entry
- **Tenant isolation:** Log segmentation ensures data integrity across different school departments
- **Audit logging:** Complete trail of all actions for accountability and troubleshooting

Results & Impact

100%	4	0
Request Visibility	Priority Levels	Lost Tickets

- Eliminated lost or forgotten support requests through centralized tracking
- Enabled IT staff to prioritize effectively with visual urgency indicators
- Reduced follow-up inquiries with automatic email notifications
- Created documentation trail for identifying and addressing recurring issues

- Delivered a professional, branded experience that staff actually want to use

Technology Stack

HTML5 | CSS3 | JavaScript | RESTful API | Database Backend | Email Integration | Rate Limiting | Responsive Design

Conclusion

This project demonstrates the ability to understand the unique needs of educational institutions and deliver practical, user-friendly solutions. By focusing on the actual workflows of school IT departments—rather than forcing a generic enterprise tool—the system achieved immediate adoption and measurable improvements in support request handling. The custom branding and accessibility features show attention to the details that make technology welcoming rather than intimidating for non-technical users.